BAHRIA UNIVERSITY, (Karachi Campus)

Department of Software Engineering

Assignment 2 - Spring 2025

COURSE TITLE:	DevOps	COURSE CODE:	SEN-429
Class:	**BSE-8 A & B**	Shift:	**Morning**
Course Instructor:	Sohaib Ur Rehman	Time Allowed:	**1 Week**
Submission Date:	**18th May 2025**	Max. Marks:	**5 Marks**

Assignment Title: Deploying Two NGINX Containers in a Single ECS Task with Load Balancer**[CLO4: 5 Marks]**

Assignment: Deploy Two NGINX Containers on ECS with Load Balancer Using a Single Task Definition

Overview:

In this assignment, you are required to deploy two containers running the NGINX web server using **Amazon Elastic Container Service (ECS)** with the **Fargate launch type**. You will use a **single task definition** to define both containers and configure an **Application Load Balancer (ALB)** to expose them to the internet.

The deployment will be carried out in **two phases**:

- Initially, the ECS service will launch **only one container** (nginx1) and attach it to the load balancer.
- You will then **modify the ECS service** configuration to include the **second container** (nginx2) from the same task definition, without creating a new task definition or scaling the number of tasks.

This assignment reinforces your understanding of ECS architecture, container orchestration, load balancing, and service modification workflows.

Deliverables:

You must submit the following items:

1. **Task Definition Screenshot** showing both containers (`nginx1` and `nginx2`) in a single definition.
2. **ECS Service Screenshot (Before and After Modification)**:
   - Before: Service configured with only `nginx1`.
   - After: Service includes both `nginx1` and `nginx2`.
3. **Running Task Screenshot** displaying both containers within the same task.
4. **Load Balancer DNS Access Proof**:
   - Access to `nginx1` container confirmed (e.g., browser screenshot).
   - Access to `nginx2` container confirmed after service modification.
5. **Brief Written Summary** (1-2 paragraphs) describing:
   - How the service was initially deployed.
   - What changes were made to include the second container.
   - Any challenges encountered and how they were resolved.